



Reducing Nitrous Oxide Waste in Healthcare Settings

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Background

- Nitrous oxide (N₂O) is a widely used anesthetic gas with a global warming potential ~265–300× that of CO₂.
- Despite declining clinical use, N₂O emissions remain high due to unnoticed or unaddressed infrastructure inefficiencies.
- At UHealth Tower, an internal review raised concern about environmental impact stemming from centralized gas delivery.
- Identifying and reducing anesthetic gas waste is a key target in healthcare sustainability efforts.

Aim

Primary Aim: Reduce N₂O emissions and waste by transitioning from piped manifold systems to portable E-cylinder use.

Objectives

- Quantify current N₂O loss from manifold leaks
- Evaluate the sustainability of transitioning to E-cylinder use
- Promote awareness and accountability in OR gas use

Methods

- Retrospective analysis of N₂O purchased vs. patient use from hospital EMR and central gas supply records.
- Leak volume estimates based on monthly manifold turnover data.
- Intervention: Proposed switch to E-cylinders to eliminate manifold-based losses.

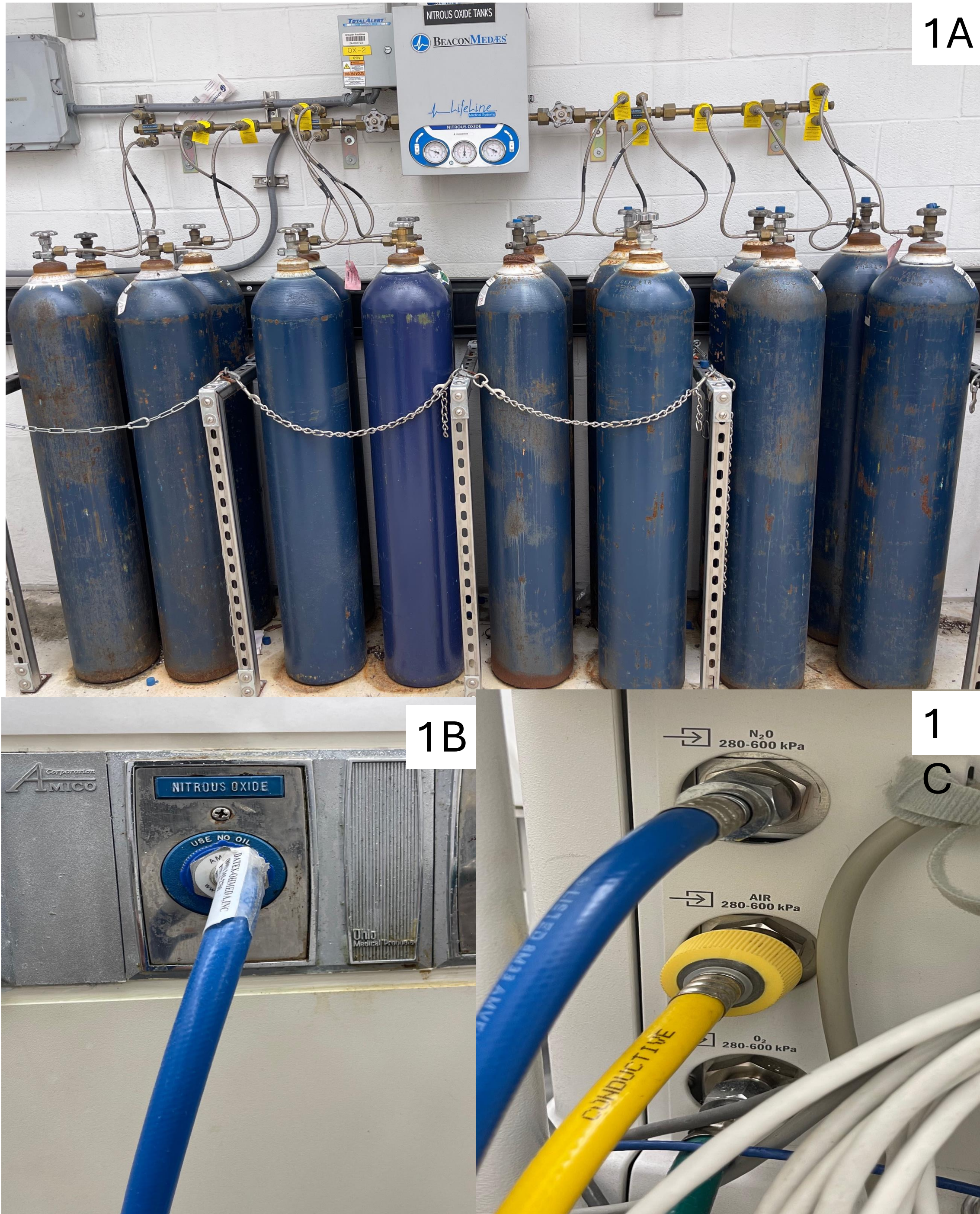


Figure 1. This figure illustrates the central manifold delivery system for nitrous oxide. Panel 1a: The 16-tank manifold at UHealth Tower, which consumes approximately eight H-tanks per month. Panel 1b: The connection from the central manifold system to the operating room wall outlet. Panel 1c: Blue tubing extending from the wall outlet to the anesthesia machine, completing the delivery path for nitrous oxide.



Figure 2. E-cylinder of nitrous oxide directly connected to the anesthesia machine for targeted, on-demand delivery.

Feature	Manifold (Central Pipeline) Delivery	E-Cylinder Delivery
Delivery Method	Central source distributed via hospital pipeline	Directly attached to anesthesia machine
Targeted Use	No – pressurized system connected to all ORs	Yes – only delivers to active ORs
Leak Risk	High (system-wide leaks difficult to detect)	Low (localized, can be isolated easily)
Gas Waste	Substantial (continuous pressurization causes waste)	Minimal
Monitoring/Control	Centralized; leaks can go unnoticed	Localized and easy to track
Installation Cost	High (infrastructure + maintenance)	Low
Operational Cost	High due to ongoing gas loss	Cost-effective over time
Clinical Flexibility	Fixed – relies on building infrastructure	High – portable and adaptable per room
Maintenance Complexity	Complex – requires entire system checks	Simple – individual cylinder management

Results

- At UHealth Tower, only an average of 6.4% of purchased N₂O is used clinically; 93.6% is lost via leakage and inefficient delivery
- Leak confirmed** by disconnecting the manifold from all ORs: H-cylinder levels continued to fall despite no clinical use.
- Monthly average loss: **108,000 L of N₂O**, translating to **~58,000 kg CO₂e of environmental waste**.

Conclusion

- Manifold leaks can cause **over 90% of purchased N₂O to be wasted**, leading to substantial unnecessary greenhouse gas emissions.
- Quantifying this loss demonstrates a critical **opportunity for targeted quality improvement** in anesthetic gas delivery systems.
- Transitioning to **portable E-cylinder systems** can **reduce N₂O waste by >99%**, yielding significant environmental and cost benefits.
- This intervention supports institutional goals for **sustainability and climate responsibility** within healthcare.

Quality Improvement Plan

- Phased implementation** of E-cylinder use across operating rooms.
- Continuous monitoring** of N₂O procurement and consumption to track waste reduction.
- Staff education and engagement** to promote sustainable anesthesia practices.

References

